

COURSE NAME: Fundamentals Of Data Science

COURSE CODE: DSA0404

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Unit 3: Hands on Exercises for Variance, Sampling, Covariance and Correlation

1. The delivery times of five online orders are: 20, 25, 30, 35, 40 minutes

Calculate the mean and population variance.

Mean:

$$\bar{x} = \frac{20 + 25 + 30 + 35 + 40}{5} = 30$$

Population variance:

$$\begin{aligned}\sigma^2 &= \frac{(20 - 30)^2 + (25 - 30)^2 + (30 - 30)^2 + (35 - 30)^2 + (40 - 30)^2}{5} \\ &= \frac{100 + 25 + 0 + 25 + 100}{5} = \boxed{50}\end{aligned}$$

2. A university has 5,000 students. A random sample of 100 students is selected to estimate the average daily study time. Explain why sampling is used instead of surveying all 5,000 students

- It saves **time and cost**.
- It requires less effort to collect data.
- A properly selected random sample can provide a **reliable estimate** of the whole population.

Answer: Sampling is more efficient and practical than surveying all 5,000 students.

3. From a sample of 200 customers, 150 customers are satisfied with a product. Calculate the sample proportion of satisfied customers and express it as a percentage.

Satisfied customers = 150

Total customers = 200

$$\hat{p} = \frac{150}{200} = 0.75$$

Percentage:

$$0.75 \times 100 = \boxed{75\%}$$

Answer: Sample proportion = 0.75 or 75%

4. Calculate the covariance between the following variables:

X	Y
2	5
4	7
6	9
8	11

Determine whether the relationship is positive or negative.

Mean of X = 5, Mean of Y = 8.

Using population covariance:

$$\begin{aligned} Cov(X, Y) &= \frac{\sum (X - \bar{X})(Y - \bar{Y})}{n} \\ &= \frac{9 + 1 + 1 + 9}{4} = \boxed{5} \end{aligned}$$

Since covariance is **positive**, X and Y have a **positive relationship**.

5. The following data represents hours of exercise (X) and calories burned (Y):

X	Y
1	200
2	300
3	400
4	500
5	600

Calculate the covariance and interpret the result.

MeanX=3

Mean Y = 400

$$\begin{aligned} Cov(X, Y) &= \frac{(-2)(-200) + (-1)(-100) + (0)(0) + (1)(100) + (2)(200)}{5} \\ &= \frac{400 + 100 + 0 + 100 + 400}{5} = \boxed{200} \end{aligned}$$

Interpretation: The positive covariance shows that as exercise hours increase, calories burned also increase.

6. Calculate Pearson's correlation coefficient for the following dataset and interpret the strength and direction of the relationship:

Study Hours (X)	Marks (Y)
1	45
2	50
3	55
4	65
5	70

Formula:

$$r = \frac{n \sum XY - \sum X \sum Y}{\sqrt{[n \sum X^2 - (\sum X)^2][n \sum Y^2 - (\sum Y)^2]}}$$

Substituting the values:

$$\begin{aligned} r &= \frac{(5)(905) - (15)(285)}{\sqrt{[(5)(55) - (15)^2][(5)(16575) - (285)^2]}} \\ &= \frac{4525 - 4275}{\sqrt{[275 - 225][82875 - 81225]}} \\ &= \frac{250}{\sqrt{(50)(1650)}} \\ &= \frac{250}{\sqrt{82500}} \\ &= \frac{250}{287.23} \\ &\boxed{r \approx 0.87} \end{aligned}$$

Interpretation: There is a strong positive correlation between study hours and marks.